

SmartRoad - Traffic Accident Analytics

Automated Analytical Report

Generated on 01 September 2026, 16:38 from the configured collision dataset.

This report is generated directly by the SmartRoad dashboard as an additional automation feature. The predictive target is high-severity collision (Fatal or Serious) versus Slight.

1. Dataset Overview

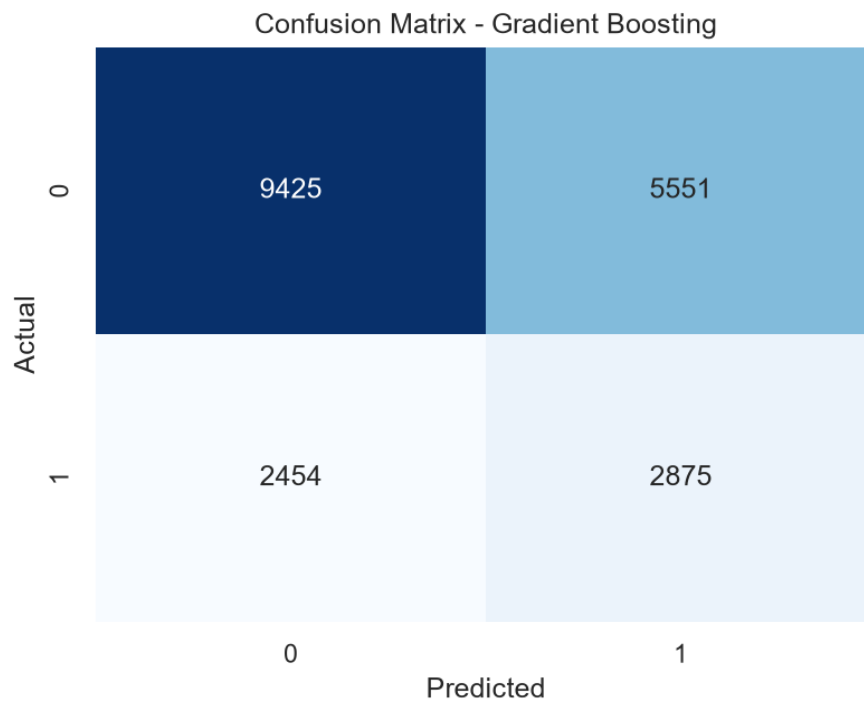
Metric	Value
Raw rows	101,525
Clean rows	101,525
Selected model	Gradient Boosting
Positive/high-severity rate	26.24%
Training records	81,220
Test records	20,305

2. Model Performance

Model	Accuracy	Precision	Recall	F1	CV F1
Random Forest	65.01%	33.85%	34.92%	34.38%	34.59%
Logistic Regression	57.33%	32.60%	58.66%	41.91%	41.85%
Gradient Boosting	60.58%	34.12%	53.95%	41.80%	42.09%

Selected model: **Gradient Boosting**, based on the highest mean 5-fold cross-validated F1-score.

3. Confusion Matrix

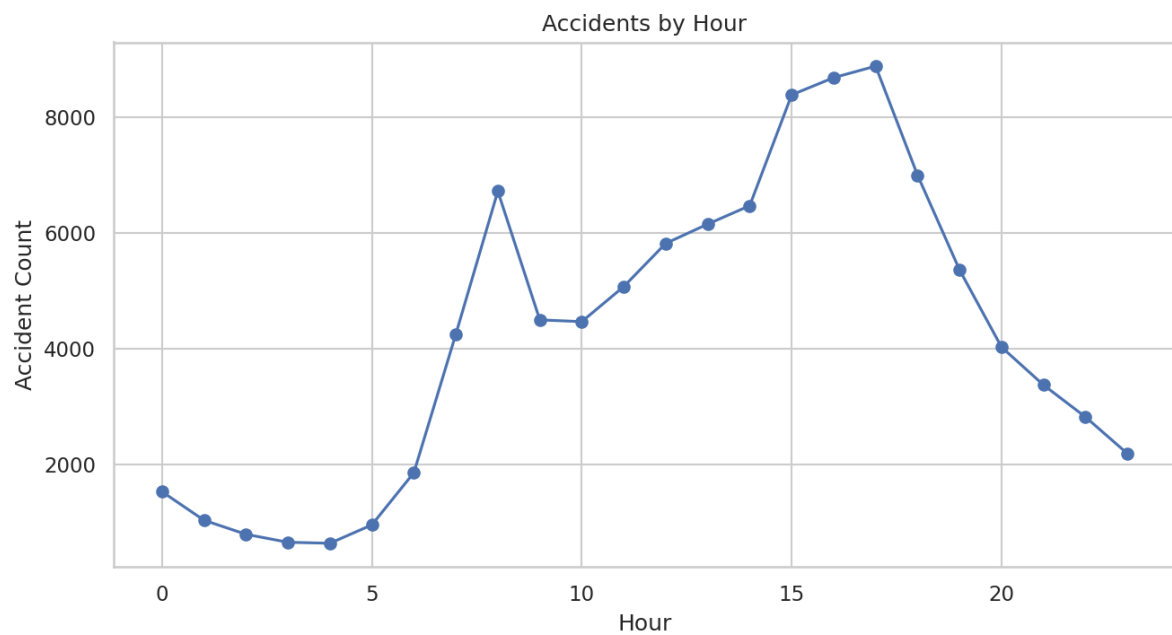


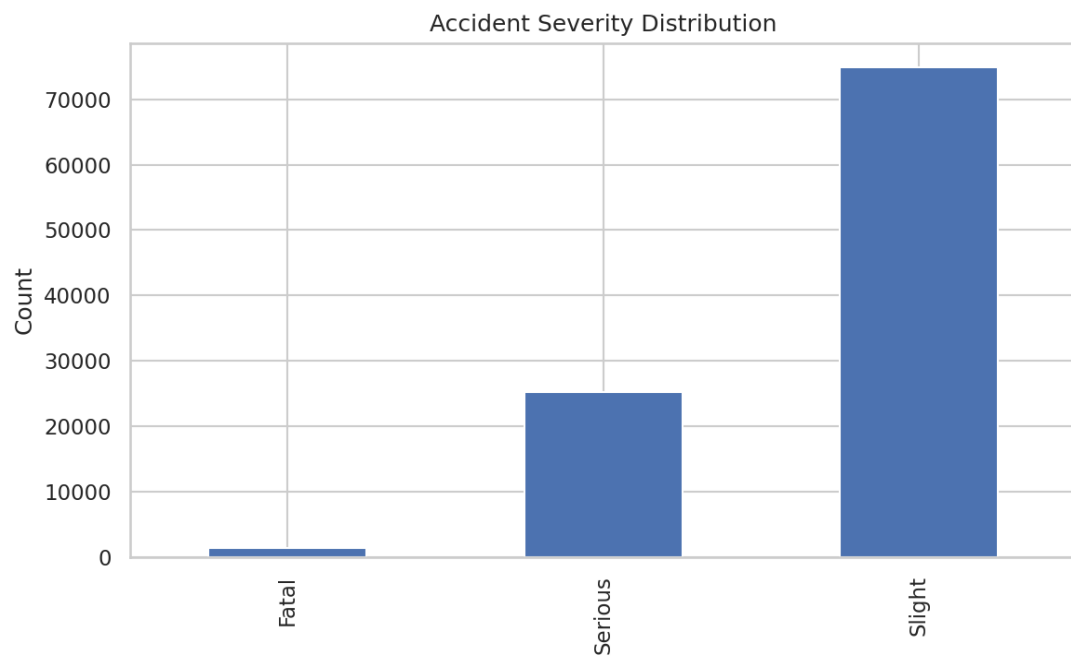
Held-out test metrics for the selected model: accuracy 60.58%, precision 34.12%, recall 53.95%, and F1 41.80%.

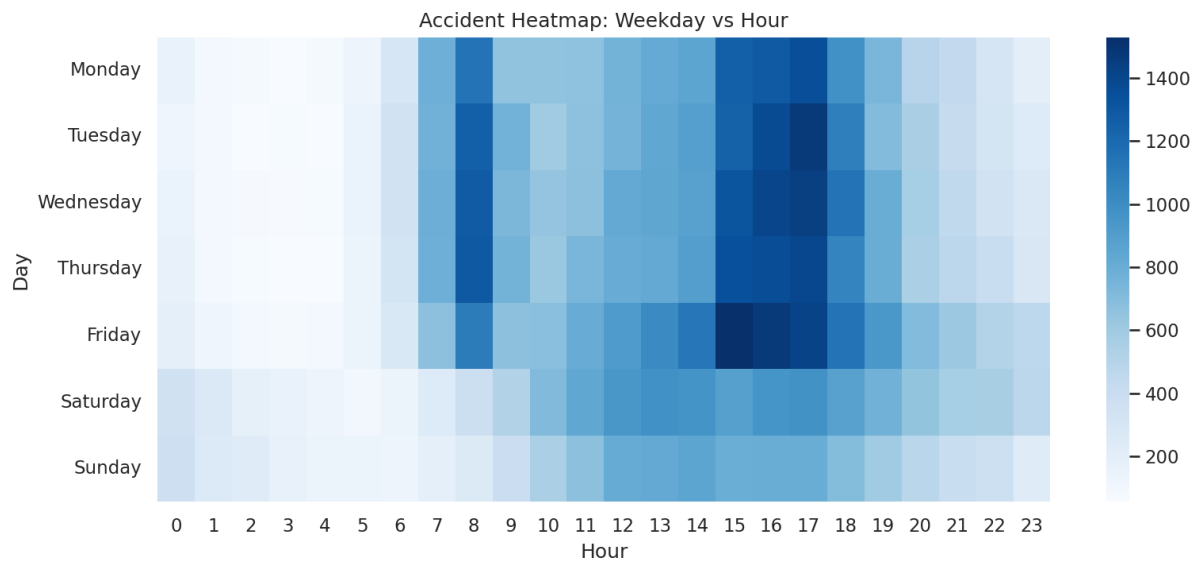
4. Key Features

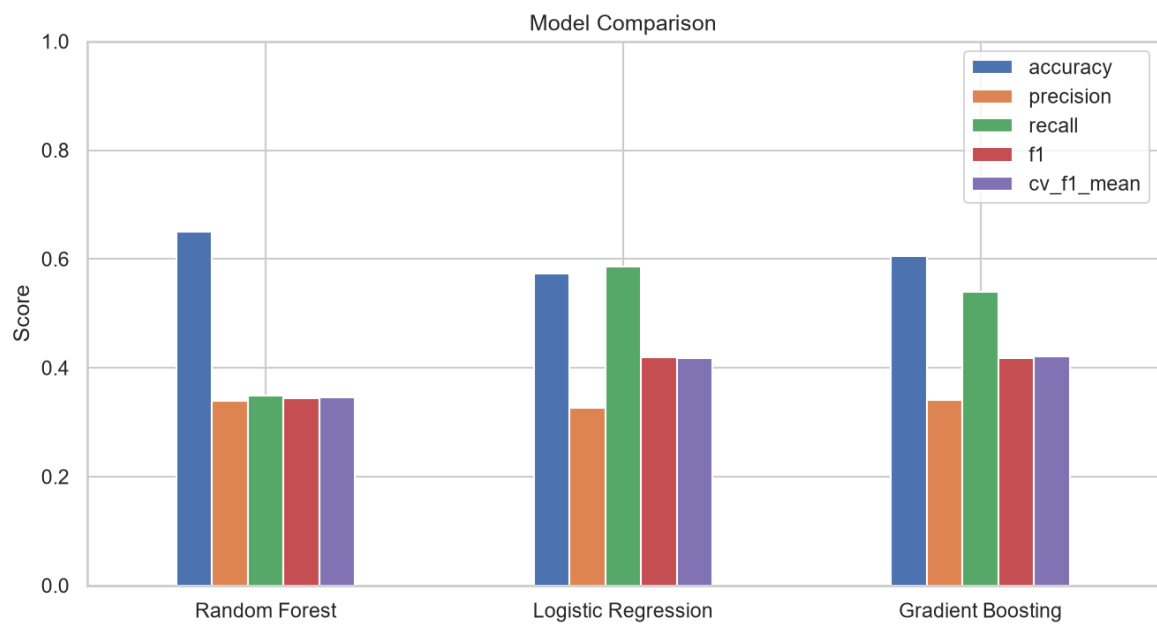
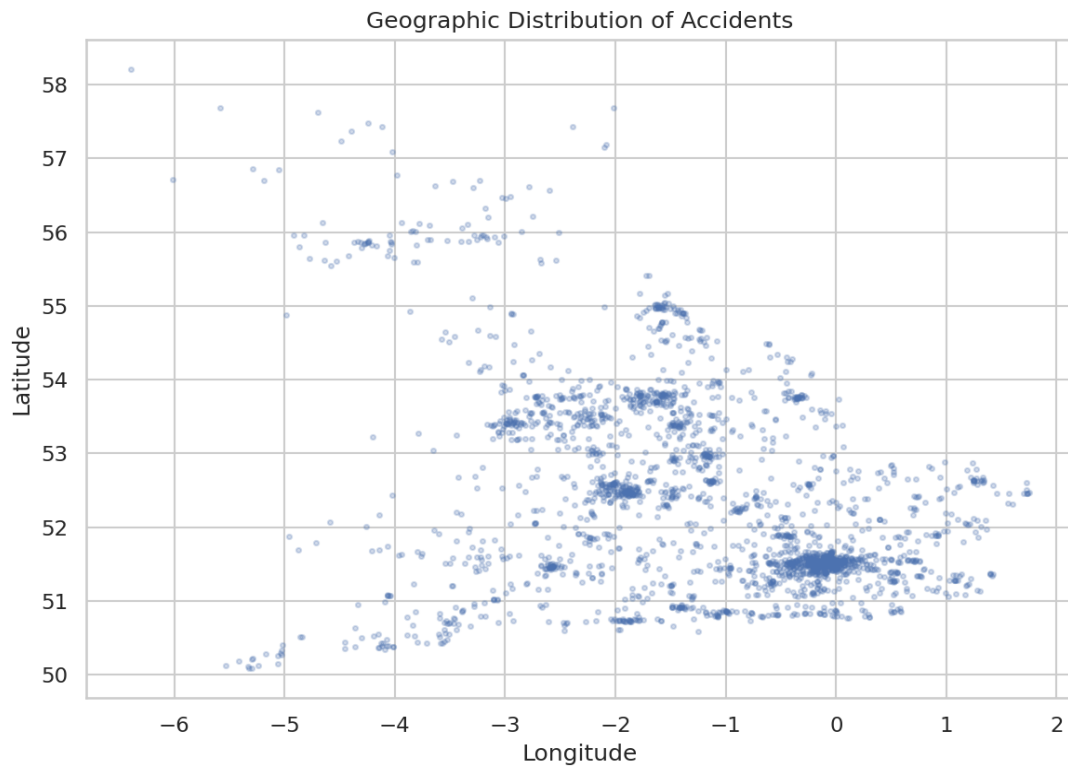
Feature	Mean importance	Std. deviation
number_of_vehicles	0.04165	0.00301
urban_or_rural_area	0.01993	0.00151
speed_limit	0.01917	0.00197
road_type	0.01271	0.00138
hour	0.00727	0.00189
weather_conditions	0.00539	0.00136
first_road_class	0.00371	0.00084
junction_detail	0.00338	0.00058
month	0.00261	0.00082
night	0.00187	0.00052

6. Visual Analytics









7. Conclusion

SmartRoad combines preprocessing, exploratory analytics, predictive classification, model comparison, cross-validation, confusion-matrix evaluation, geographic hotspot clustering and interactive prediction. This automatically generated report packages the current analytical results for review and documentation.